

Vaibhav Singh

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SUMMARY

Innovative Computer Science Engineering student driven by a passion for building impactful, real-world software solutions. Eager to apply strong analytical problem-solving skills to tackle complex challenges.

EDUCATION

SRM UNIVERSITY, RAMAPURAM

B.Tech · Computer Science and Engineering
Chennai, India

June 2023 – Present

GPA: 9.2/10.0

ST. PAUL'S COLLEGE

12th ISC Board – 78%

10th ICSE Board – 90%

April 2021 – May 2022

April 2019 – May 2020

EXPERIENCE

Technical Team Member

April 2024 – present

INTELLECTS SRMIST Ramapuram

- Collaborated within a cross-functional team to architect and deploy scalable full-stack web applications utilizing Java, Spring Boot, and React.js.
- Spearheaded a campus-wide competitive programming initiative, successfully growing and mentoring an active community of 100+ student participants.
- Designed and delivered technical seminars for online campus events, engaging audiences of 65+ attendees to foster software development skills.

PROJECTS

NES Emulator

[Link](#)

- Engineered a cycle-accurate emulation of the MOS 6502 CPU, precisely implementing official instruction decoding, status flags, and complex addressing modes to match hardware timing constraints.
- Built a robust Picture Processing Unit (PPU) featuring scanline-by-scanline rendering, background tile generation, OAM sprite evaluation, and accurate fine-scrolling register behavior.
- Implemented dynamic memory bank switching by adding support for common iNES mappers (MMC1, MMC3), drastically expanding compatibility to run advanced commercial ROMs.

3D Raycasting Engine

[Link](#)

- Developed a 3D graphics engine from scratch using Java, implementing the Digital Differential Analyzer (DDA) algorithm to render a 3D perspective from a 2D grid.
- Applied linear algebra and vector mathematics to calculate precise ray-wall intersections and correct camera plane fish-eye distortion.
- Optimized the rendering loop and rasterization logic to maintain a consistent 60+ FPS with real-time collision detection and frame-independent movement.

HTTP Server

[Link](#)

- Engineered a custom, multithreaded HTTP/1.1 web server from scratch using core Java (Java Sockets, I/O Streams) to process client-server communication over TCP/IP.
- A robust request parser capable of interpreting HTTP methods (GET, POST), headers, and payloads, ensuring strict adherence to HTTP protocol standards.
- Implemented a thread-pool architecture using Java ExecutorService to manage concurrent client connections, preventing server blocking and optimizing resource utilization.

SKILLS

Programming Languages:	Java, Python
Languages and Frameworks:	Spring Boot, Java Web Toolkit, Vaadin
Tools and Technologies:	Git, Firebase, MongoDB, Maven, Docker
Areas of Interest:	Object-Oriented Programming, Problem Solving, Backend Development, REST APIs

CERTIFICATIONS

Oracle Certification Oracle Certification for DevOps	Link Oct 2025
Unstop Certification Unstop 30 Days DSA bootcamp in Java	Link Jun 2025
AWS Training & Certification AWS Foundation AWS Training Certification for Machine Learning Basics	Link Aug 2025
NPTEL Certification NPTEL - NPTEL24CS96S156301331 NPTEL Certification for Data Structures and Algorithms in Java	Link Oct 2024
SIH 2025 SRMIST RAMAPURAM	Link Sept 2025